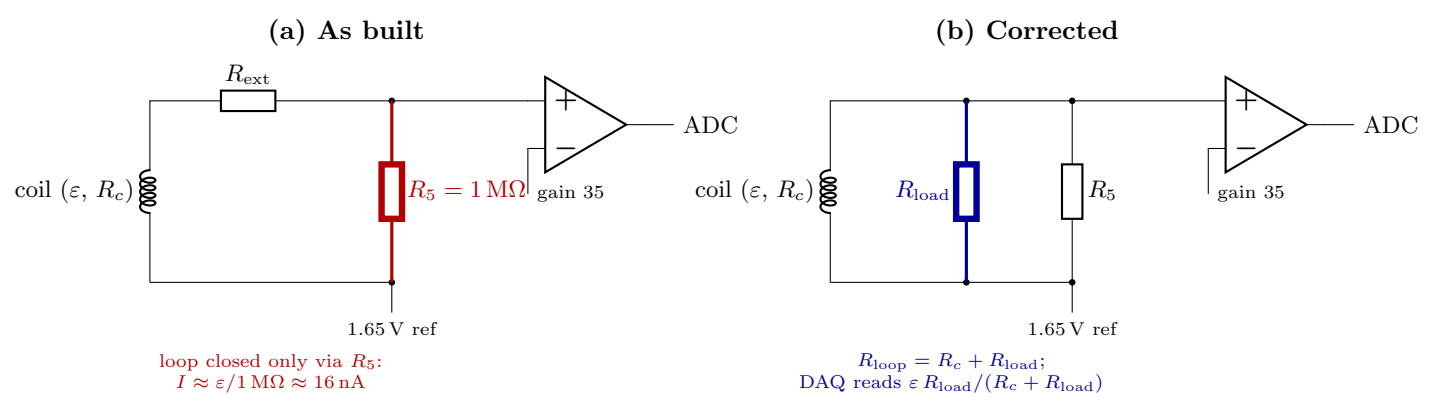

(a) As built
R_{ext}
coil (ε, R_c)
$R_5 = 1\,\text{M}\Omega$
gain 35
ADC
1.65 V ref
loop closed only via R_5:
$I \approx \varepsilon/1\,\text{M}\Omega \approx 16\,\text{nA}$
(b) Corrected
coil (ε, R_c)
R_{load}
R_5
gain 35
ADC
1.65 V ref
$R_{\text{loop}} = R_c + R_{\text{load}}$;
DAQ reads $\varepsilon\, R_{\text{load}}/(R_c + R_{\text{load}})$